

Md. Mohiuddin

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Education

Bachelor of Science (BSc) in Physics

University of Dhaka, Bangladesh

GPA: **3.15/4.00**



Research Interests

- **Computational optoelectronic materials (perovskites & related semiconductors):** DFT (VASP/HSE06), DFPT/Phonopy, BoltzTraP2 → device-level drift-diffusion modeling for photodiodes/PV.
- **Defect physics & doping control:** formation energies, charge-state thermodynamics, nonradiative losses, and doping limits that set transport/quantum efficiency.
- **Stability & synthesis windows:** convex-hull distance, phonon/elastic criteria, and ab-initio MD to de-risk experimental growth.
- **ML-assisted discovery:** pymatgen/MP-API descriptors with uncertainty-aware screening (scikit-learn/XGBoost) to prioritize costly DFT.
- **Synthesis of functional nanomaterials (MOFs & heterostructures):** solvothermal/hydrothermal routes, core-shell engineering (e.g., **MIL-100(Fe)@TiO₂**), and thin-film deposition; characterization by XRD, SEM/EDS, TEM, BET, UV-Vis, FTIR.
- **Photocatalysis & band-engineering:** kinetic analysis (pseudo-first-order k , AQE, TOF), reusability/cycling tests, and DFT-guided band alignment (type-II/Schottky) to link structure → activity.



Publications & Preprints

- **Stability Analysis and Optoelectronic Properties of Mg₃ZBr₃ (Z = As, Sb, Bi) Perovskites for Evaluating the Performance in PIN Photo Diode**
First Author; Supervisor: Dr. Alamgir Kabir
Published in ACS Omega. [🔗 DOI](#)
- **Sr₂PrSbO₆ Double Perovskite as a Solar-Blind UV Photodetector: A Combined DFT and Device Simulation Study**
First Author; Supervisor: Dr. Alamgir Kabir
Accepted for publication in Journal of Materials Chemistry C (RSC); In Press.
- **Interface Band Offsets as the Common Limit Across Four Lead-Free Perovskite Solar-Cell Absorbers: A Coupled Opto-Electrical COMSOL Study**
First Author; Supervisor: Dr. Alamgir Kabir
Under Revision in Advanced Electronic Materials (Wiley).
- **Conformal Prediction Quantifies the Reliability Limits of Machine-Learned Band-Gap Screening in Oxide Double Perovskites**
First Author; Supervisor: Dr. Alamgir Kabir
Under Review in Machine Learning: Science and Technology (IOP Publishing).

Ongoing Projects:

- Memristor device performance of ScAlN using COMSOL under the supervision of Prof. Dr. Alamgir Kabir.
- Radiation effects on $\text{Sr}_2\text{PrSbO}_6$ double-perovskite PIN diodes using COMSOL, under the supervision of Prof. Dr. Alamgir Kabir.
- Synthesis of $\text{TiO}_2@\text{MIL-100 (Fe)}$ microspheres and its photocatalytic application under the supervision of Prof. Dr. Mahabub Alam Bhuiyan



Research Experience

Undergraduate Research Assistant, Computational Materials Science Laboratory Jan 2024

– Present

Supervisor: Dr. Alamgir Kabir

- Performed first-principles (DFT) calculations on halide perovskites Mg_3ZBr_3 ($\text{Z} = \text{As, Sb, Bi}$) to evaluate structural stability, phonon spectra, and optoelectronic properties for PIN photodiodes.
- Carried out HSE06 band-structure and phonon calculations; identified Sb/Bi compositions in the optimal 1.5–1.7 eV band-gap window and analyzed soft phonon modes as a guide for composition tuning.
- Linked materials properties to device performance via PIN drift–diffusion simulations, connecting band gaps to responsivity onset.
- Simulated memristor device performance of ScAlN using COMSOL Multiphysics, including I–V characteristics and switching behavior under different biasing conditions.
- Mentored 4 junior students on DFT workflows (input preparation, convergence tests, post-processing) and basic data analysis/visualization.

Undergraduate Research Assistant, Semiconductor Technology Research Centre (STRC)

Oct 2025 – Present

Supervisor: Prof. Dr. Mahabub Alam Bhuiyan

- Participating in the synthesis of $\text{TiO}_2@\text{MIL-100(Fe)}$ microspheres and evaluating their photocatalytic performance through dye-degradation and related optical measurements.
- Providing research guidance and mentoring to 2 junior students on experimental protocols, data recording, and interpretation of results.



Technical Skills

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- **Programming:** Python (Pandas, NumPy, PyTorch, Scikit-learn, Pymatgen), MATLAB, C++
 - **Software:** VASP, CASTEP, Phonopy, BoltzTrap, COMSOL, OriginLab
 - **Data Analysis:** Supervised Learning (KNN, Linear Regression, Ridge Regression, Lasso, Logistic Regression, XGBoost, Feature Engineering, Hyperparameter Tuning), Unsupervised Learning with PyTorch
 - **Operating Systems:** Windows, Linux, MacOS



Conference Poster Presentation

Md. Mohiuddin, *A DFT Study of $\text{Sr}_2\text{PrSbO}_6$* (Poster). Centennial Celebration of Bose–Einstein Statistics, 2024. [Zenodo DOI: 10.5281/zenodo.17409885](https://zenodo.org/doi/10.5281/zenodo.17409885)